

Task 1: Log in to Cisco Packet Tracer or any network simulation tool, add a switch, and create two VLANs named 'Sales' and 'Support' with VLAN IDs 10 and 20.

Vlan is creating multiple separate virtual networks within a single Switch.

Even if devices are plugged into the exact same switch sitting on the same desk, putting them into different VLANs makes them act as if they are connected to entirely separate hardware in different rooms.

In the Switch,

Go to CLI

switch> enable

Switch # config terminal

Switch(config)# vlan 10

Switch(config-vlan)# name Sales

Switch(config-vlan)# exit

Switch(config)# vlan 20

Switch(config)# name Support

Switch(config)# exit

Switch(config)# do wr

```
% Invalid input detected at '^' marker.

Switch(config)#vlan 10
Switch(config-vlan)#name Sales
Switch(config-vlan)#exit
Switch(config)#clan 20
^
% Invalid input detected at '^' marker.

Switch(config)#vlan 20
Switch(config-vlan)#name Support
Switch(config-vlan)#exit
Switch(config)#do wr
Building configuration...
[OK]
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
10	Sales	active	
20	Support	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

Switch#

Task 2: Assign three different switch ports to VLAN 10 (Sales) and two ports to VLAN 20 (Support), then verify the VLAN membership using the show vlan command. Hint: Use the interface range command to select multiple ports efficiently.

```

switch> enable
Switch # config terminal
Switch(config-if-range)# interface range fa0/1 - 3
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10
Switch(config-if-range)# exit
Switch(config-if-range)# interface range fa0/4 - 5
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 20
Switch(config-if-range)# exit

```

```
Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24, Gig0/1 Gig0/2
10	Sales	active	Fa0/1, Fa0/2, Fa0/3
20	Support	active	Fa0/4, Fa0/5
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```

Switch#
Switch#

```

Task 3: Connect two routers using a serial or Ethernet link in your simulation tool, assign IP addresses to their interfaces, and configure a static route on Router 1 to reach the LAN behind Router 2.

Addressing Plan:

- **R1 & R2 WAN Link:** Network address: 10.0.0.0/30
 - R1 Gi0/0: 10.0.0.1 255.255.255.0
 - R2 Gi0/0: 10.0.0.2 255.255.255.0

- **LAN behind Router 2:** Network 192.168.20.0/24
 - R2 Gi0/1 (Gateway): 192.168.20.1 255.255.255.0



```

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/24 is directly connected, GigabitEthernet0/0
L       10.0.0.1/32 is directly connected, GigabitEthernet0/0
S       192.168.20.0/24 [1/0] via 10.0.0.2

```

Task 4: Simulate a scenario where a device in the 'Sales' VLAN tries to ping a device in the 'Support' VLAN and explain why the ping fails, then suggest what configuration would allow inter-VLAN communication. Hint: Think about router-on-a-stick or Layer 3 switch concepts.

Ping fail because of 2 reason

Layer 2 Isolation (Broadcast Domains)

A switch treats each VLAN as a completely separate, isolated logical network (a distinct **Layer 2 broadcast domain**).

- When a PC in VLAN 10 sends a frame, the switch will **never** forward that frame directly to a port assigned to VLAN 20.
- Even though the PCs are plugged into the exact same physical switch, they are logically cut off from each other.

2. Separate Subnets & Missing Layer 3 Routing

Devices in different VLANs have different IP subnets (e.g., VLAN 10 is 192.168.10.0/24 and VLAN 20 is 192.168.20.0/24).

- When PC 1 (192.168.10.10) tries to ping PC 2 (192.168.20.10), it realizes the destination IP is on a different network subnet.
- Because it's on a different subnet, PC 1 cannot ping PC 2 directly . Instead, it must send the packet to its Default Gateway (a router or Layer 3 switch) to route the packet across networks.

To make the ping work: one of the two methods can be used:

1. Router-on-a-Stick

[Apartment Switch] ===== (Single Cable) ===== [Outside Mailman]

(VLAN 10 & VLAN 20)

(Router)

- How it works:
 1. A resident in Apartment 10 drops a letter in a box outside the building.
 2. An outside mailman (the Router) drives up, picks up the letter, reads the address, and walks it back into the building to deliver it to Apartment 20.

2. Layer 3 Switch

[Apartment Switch with Built-in Mail Chute]

|— Apartment 10 Door

L— Apartment 20 Door

- How it works:
 1. The building itself is built with a high-speed, automated internal mail chute system (Layer 3 Switch).
 2. A resident drops a letter into the chute in Apartment 10, and the building automatically drops it directly into Apartment 20 inside the wall.